11-April-2026

Agenda:

statistics - 6

- ↳ covariance & correlation + demo
- ↳ probability distribution
  - ↳ Discrete
    - ↳ pmf
    - ↳ cdf
  - ↳ continuous
    - ↳ pdf
    - ↳ cdf
- ↳ ~~Distribution~~

covariance :

variance / std → how ~~all data~~ spread
one column
variable

Que: if one variable changes , does another also change ?

id		live_class_attended (100)		completion_rate
0	← ✗	1	→ ✓	≈ 0 %
1	←	100	→	≈ 100
2	←	50	→	>50

col-1 col-2 col-3 col-4 ... col-n

pattern

w.r.t → col-1 → col x, col y, col z → pattern → manully, stats live_class_attended (100)   completition_rate

↑   ↑   → directly proportional

body_weight   running speed

↑   ↓   → inversely proportional

body_weight   row-index (pandas)

no-relation

covariance → +ve → directly proportional

covariance → 0 → no relation

covariance → -ve → inversely proportional

Student	Study-hours	Marks
A	1	30
B	2	40
C	3	50
D	4	60

formula : $$\mathrm{cov}(X, Y) = \frac{\sum (x_i - \bar{x})(y_i - \bar{y})}{n-1}$$

step1:

$$\bar{x} = \frac{1+2+3+4}{4} = 2.5$$

$$\bar{y} = \frac{30+40+50+60}{4} = 45$$

Student	Study-hours	Marks	$(X-\bar{x})$	$(Y-\bar{y})$	product
A	1	30	$1-2.5 = -1.5$	$30-45 = -15$	22.5
B	2	40	$2-2.5 = -0.5$	$40-45 = -5$	2.5
C	3	50	$3-2.5 = 0.5$	$50-45 = 5$	2.5
D	4	60	$4-2.5 = 1.5$	$60-45 = 15$	22.5

$\Sigma = 50$

$= \frac{50}{3} = 16.67 \rightarrow$

+ve ✓ → directly proportional

→ ≈0

→ -ve

← covariance

Hours	Stress	$x-\bar{x}$	$y-\bar{y}$	product
1	90	-1	+20	-20
2	70	0	0	0
3	50	+1	-20	-20
				-40

$\frac{-40}{n-1} = <0 =$ negative

ML →

$\boxed{\text{20 columns}}$

id x y z a b c

→ no relation + $\boxed{\text{relation}}$

↓

data → ML → formula?

issue : ?

A B

$\boxed{1 \quad 1500 \text{ mm}}$ → 1500 → 1000

$\boxed{1 \quad 70}$ → 70 → 70

$\boxed{A \quad C}$ A D → 10,000,00

A, B → unit unit

hours (mm) → large → 1000 → A - B have better relation?

A C hours marks → < large → 500

X ↓ unit

A - B → better?
A - C

$-1 \leftarrow$ range $\rightarrow 1$

A - B → 0.68
A - C → 0.78 → better

Correlation:

$-1 \leq \text{range} \leq 1$

$$\text{range} = \frac{cov(X, Y)}{\sigma_x \cdot \sigma_y}$$ → fix → normalize covariance → $\boxed{-1 \text{ to } 1}$

↓

$\boxed{\text{remove unit}}$

X	Y	$(X-\bar{x})$	$(Y-\bar{y})$	product:
1	30	-1	-10	10
2	40	0	0	0
3	50	+1	+10	10
				20

$$cov = \frac{20}{2} = 10$$

Standard Deviation:

$X: (x-\bar{x})^2: 1, 0, 1 \rightarrow \text{sum} \rightarrow 2 \rightarrow \sigma_x = \sqrt{\frac{2}{2}} = 1$

↑ variance

$Y: (Y-\bar{y})^2: 100, 0, 100 \rightarrow \text{sum} \rightarrow 200 \rightarrow \sigma_y = \sqrt{\frac{200}{2}} = 10$

$$\text{correlation}: \frac{cov(X, Y)}{\sigma_x \cdot \sigma_y} = \frac{10}{1 \cdot 10} = 1$$ → perfect

↓

directly proportion

x	y
1 ←	10
2 ←	20
3	30

x x → 1
x y → 1
y x → 1
y y → 1

$x \to 0$
$y = 1$ → $\begin{bmatrix} 00 & 01 \\ 10 & 11 \end{bmatrix}$

$\begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix} \leftarrow \begin{bmatrix} xx & xy \\ yx & yy \end{bmatrix}$

(A) ML → all column (10) (exclude index) → 9 → 86%

(B) ML → filter more → top 5 column → ML → 92%

(C) ML → top 3 → ML → 91.6%

simpler ↓ (C)

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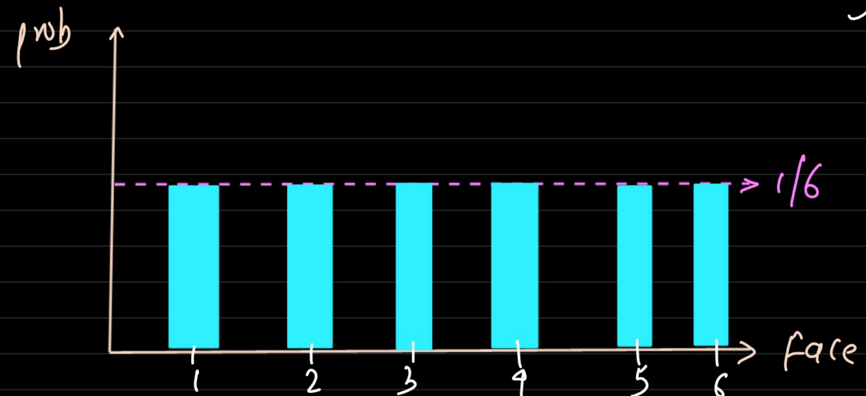
probability distribution

PMF (probability Mass Function)

> → PMF is a function, not just a plot

why?

→ To answer: "what's the chance of X happening exactly?"



> sum of all prob = 1
> only works for discrete outcomes

$S = \{1, 2, 3, 4, 5, 6\}$

$\{$
1: 1/6
2: 1/6
3: 1/6
4: 1/6
5: 1/6
6: 1/6
$\}$

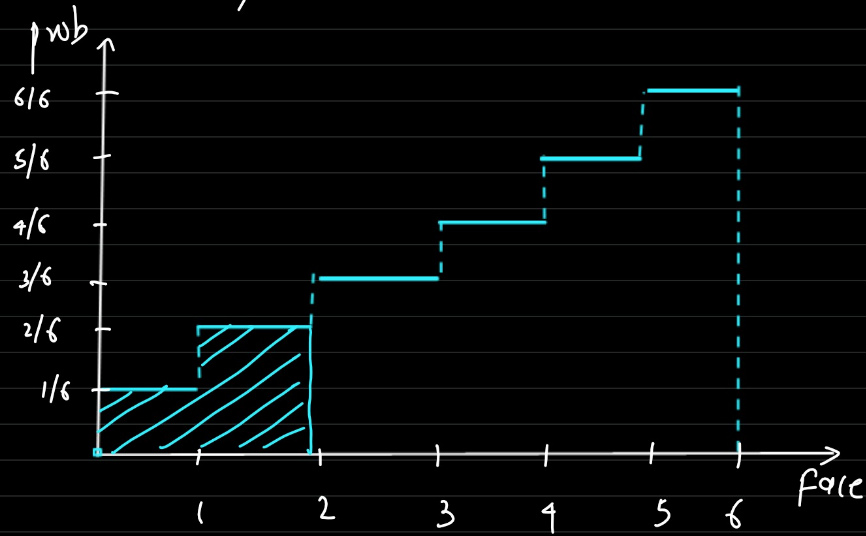
```
def pmf_dice(face):
    if face in [1,2,3,4,5,6]:
        return 1/6
    else :
        return 0
```

CDF (Cumulative Distribution Function)

→ CDF works for both discrete & continuous data
→ It gives the probability that X is $\leq$ a value

why ?
To answer : "what's the chance of X being less than or equal to 5 ?"
Example : "what the prob of rolling ≤ 5 on a dice ?"

$P(1) \to 1/6$
$P(2) \to 1/6$
$P(3) \to 1/6$
$P(4) \to 1/6$
$P(5) \to 1/6$
$P(6) \to 1/6$



$P(\leq 2) \to P(1) + P(2)$

cdf (≤ 2)

$P(\leq 5) \to P(1) + P(2) + P(3) + P(4) + P(5)$

$$p(0) = 0$$

$$p(\leq 1) = pmf(0) + pmf(1)$$

$$= 0 + 1/6$$

$$p(\leq 2) = pmf(0) + pmf(1) + pmf(2) \rightarrow 2/6$$

$\vdots$

$$p(\leq 6) = pmf(0) + \dots pmf(6) \rightarrow 6/6 \rightarrow 1$$

→ CDF always starts at 0 and end at 1

→ Discrete data, its a step function.

→ For continuous data, its a smooth curve.

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discrete → pmf
→ cdf

continuous → pmf

→ why not pmf for continous data?

→ what's the prob a student height is 160 cm [150 cm, 170 cm]

$$p(160) = \frac{1}{\infty}$$

$$= \frac{1}{\infty} = 0$$

→ what's the prob a student height is between $\underline{160\text{ cm} \,\&\, 170\text{ cm}}$?

range

pdf:

→ pdf is for continous data (like height, weight, temp)

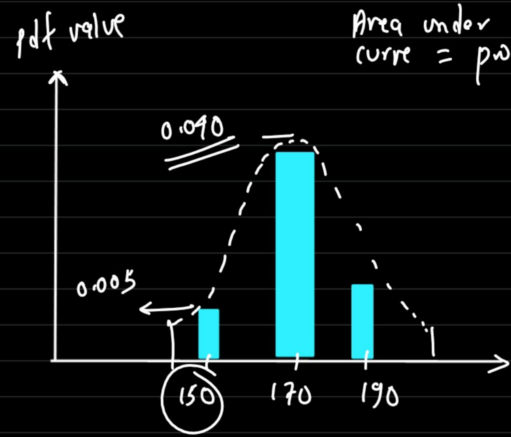
→ It shows the relative likelihood of a range of values (not exact point)

→ Continuous prob are only meaniful over intervals.

pdf as density (Not probability)

→ The height of the pdf curve at a point tells you how concentrated probabilities are around that value.

Higher pdf value = More likely to find values nearby



Area under pdf curve = probabil

Step 1: Collect Data

Height (cm) of 20 people:

$n = 20$

min – 159
max – 179

[162, 168, 171, 154, 175, 178, 165, 169, 172, 160, 167, 173, 170, 176, 164, 179, 166, 170, 174, 168]

Step 2: Create a Histogram

↓ Bins → frequency

Height (cm)	Frequency
150 - 155	1
155 - 160	0
160 - 165	3
165 - 170	6
170 - 175	6
175 - 180	4

n-bins = 6
width = 5

$$\frac{\frac{1}{20}}{5} = \frac{0.05}{5} = 0.01$$

$$\frac{1}{20 \times 5} = \frac{1}{100} = 0.01$$

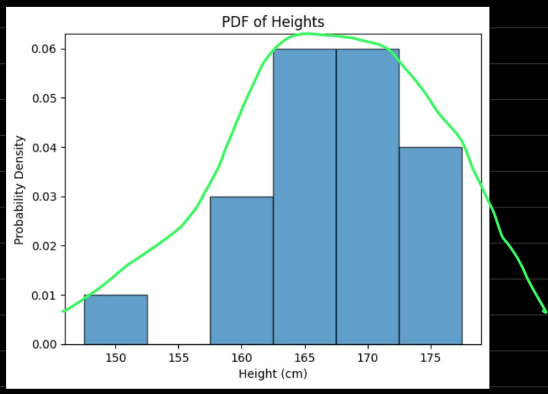
Step 3: Normalize Histogram to get pdf

pdf : probability density function

$$\text{pdf} = \frac{\frac{\text{frequency of bin}}{\text{total observation}}}{\text{Bin width}} = \frac{\text{frequency of bin}}{\text{Total observation} \times \text{Bin width}}$$

Height (cm)	Frequency	pdf
150 - 155	1	(1/20)/5 = 0.01
155 - 160	0	0
160 - 165	3	0.03
165 - 170	6	0.06
170 - 175	6	0.06
175 - 180	4	0.04

$6/20 \to 0.30$



Q: $p(165 \leq \text{height} \leq 170)$

step 4 : Calculate Area under pdf curve

→ Identify bins involved : 165 - 170 cm

→ Area = pdf value X Bin width

= 0.06 X 5 = 0.30 (30% chance)

verification :

Actual data point in 165 - 170 cm : 6 people

probability = $6/20$ = 0.30

Example 2 : $p(\text{Height} \leq 170)$

(1) Add areas upto 170 cm:

150 - 155 = 0.01 x 5 = 0.05

155 - 160 = 0 x 5 = 0

160 - 165 = 0.03 x 5 = 0.15

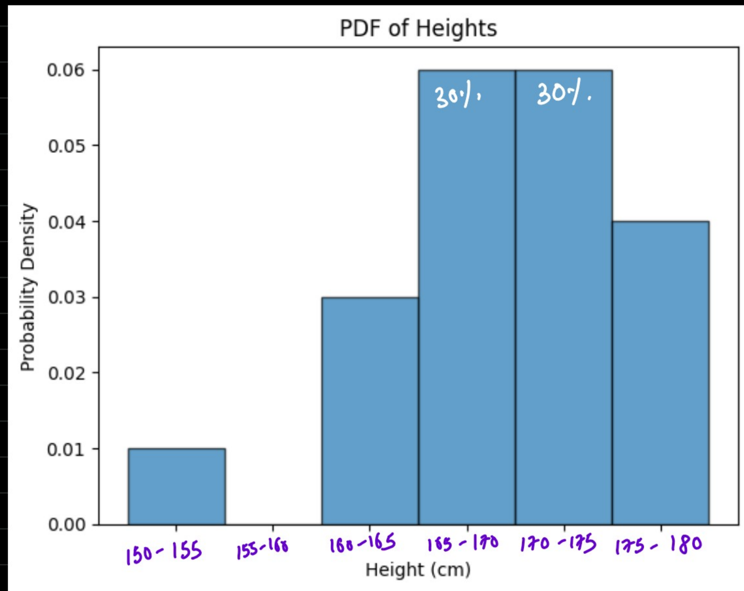
165 - 170 = 0.06 x 5 = 0.30

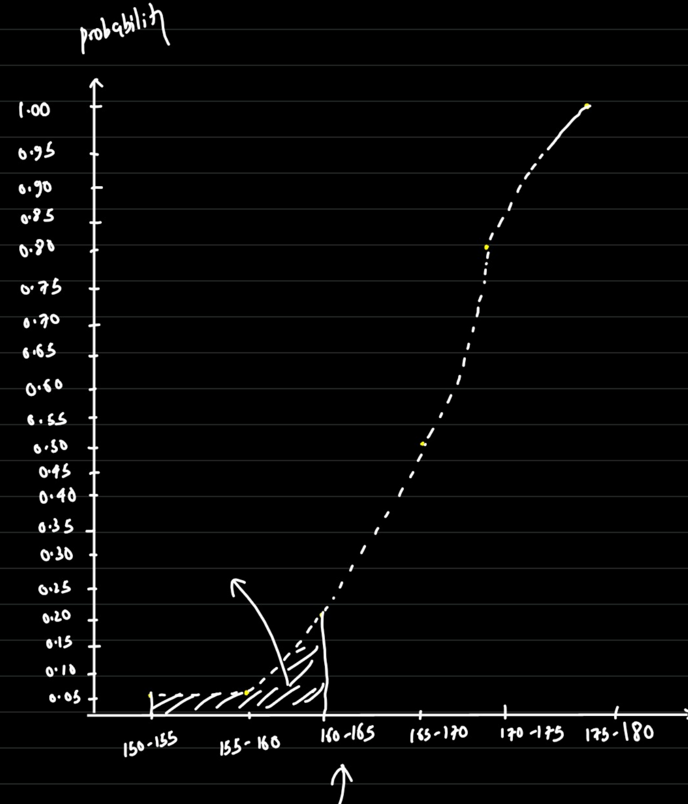
~~170 - 175~~

total = 0.05 + 0 + 0.15 + 0.30

= 0.50 → 50%

verification = $\frac{10}{20}$ = 0.5 = 50%





Height (cm)	Frequency	pdf	prob
150 - 155	1	0.01	0.05
155 - 160	0	0	0
160 - 165	3	0.03	0.15
165 - 170	6	0.06	0.30
170 - 175	6	0.06	0.30
175 - 180	4	0.04	0.20

cdf:

A → 0.05
B → 0.05 + 0 → 0.05
C → 0.05 + 0.15 → 0.20
D → 0.20 + 0.3 → 0.5
E → 0.5 + 0.3 → 0.8
F → 0.8 + 0.2 → 1

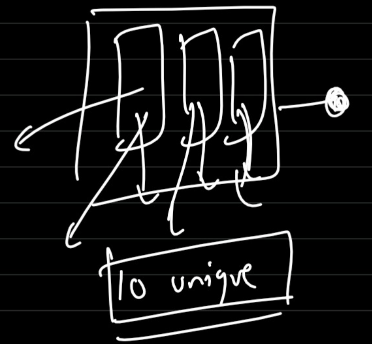
$$P(a \leq x \leq b) = CDF(b) - CDF(a)$$

$$P(165 \leq x \leq 170) = CDF(170) - CDF(165)$$

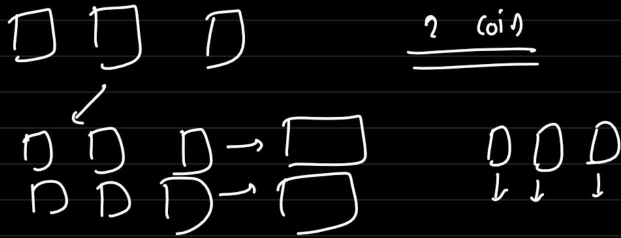
$$= P(x \leq 170) - P(x \leq 165)$$

→ lottery

10 unique



? coin



random x > 3 → 9

Random → num

fun

funct(DDD)

60%